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MONTY HALL PROBLEM



The Monty Hall Problem illustrates a surprising fact of probability. The problem is loosely based on the old game show *Let's Make a Deal* and its host, Monty Hall. In the Monty Hall Problem, you can pick one of three doors. Behind one door is a prize: a new car. Each of the other two doors opens onto a worthless goat. Say you pick Door #1. Before the door you choose is opened, the host opens another door (either #2 or #3), which leads to a goat. You can choose to either open the door you originally picked or switch to the other unopened door.

It may seem like it doesn't matter if you switch or not, but your odds do improve if you switch doors! This program demonstrates the Monty Hall problem by letting you do repeated experiments.

To understand why your odds improve, consider a version of the Monty Hall Problem with one thousand doors instead of three. You pick one door, and then the host opens 998 doors, which all reveal goats. The only two